It is easy to see that 4 and 5 as solutions for $y^2-9y+20=0$. Now taking $x^2=4$ and $x^2=5$, we get $2,-2,\sqrt{5},-\sqrt{5}$ as solutions of the given equation. ■

We note that the technique adopted above can be applied to polynomial equations like $x^6-17x^3+30=0$, $ax^{2k}+bx^k+c=0$ and in general polynomial equations of the form $a_nx^{kn}+a_{n-1}x^{k(n-1)}+\cdots+a_1x^k+a_0=0$ where k is any positive integer.

3.7.3 Zero Sum of all Coefficients

Let $P(x)=0$ be a polynomial equation such that the sum of the coefficients is zero. What actually the sum of coefficients is? The sum of coefficients is nothing but $P(1)$. The sum of all coefficients is zero means that $P(1)=0$ which says that 1 is a root of $P(x)$. The rest of the problem of solving the equation is easy.

Example 3.17

Solve the equation $x^3-3x^2-33x+35=0$.

Solution

The sum of the coefficients of the polynomial is 0. Hence 1 is a root of the polynomial. To find other roots, we divide $x^3-3x^2-33x+35$ by $x-1$ and get $x^2-2x-35$ as the quotient. Solving this we get 7 and -5 as roots. Thus $1,7,-5$ form the solution set of the given equation. ■

3.7.4 Equal Sums of Coefficients of Odd and Even Powers

Let $P(x)=0$ be a polynomial equation such that the sum of the coefficients of the odd powers and that of the even powers are equal. What does actually this mean? If a is the coefficient of an odd degree in $P(x)=0$, then the coefficient of the same odd degree in $P(-x)=0$ is $-a$. The coefficients of even degree terms of both $P(x)=0$ and $P(-x)=0$ are same. Thus the given condition implies that the sum of all coefficients of $P(-x)=0$ is zero and hence 1 is a root of $P(-x)=0$ which says that -1 is a root of $P(x)=0$. The rest of the problem of solving the equation is easy.

Example 3.18

Solve the equation $2x^3+11x^2-9x-18=0$.

Solution

We observe that the sum of the coefficients of the odd powers and that of the even powers are equal. Hence -1 is a root of the equation. To find other roots, we divide $2x^3+11x^2-9x-18$ by $x+1$ and get $2x^2+9x-18$ as the quotient. Solving this we get $\frac{3}{2}$ and -6 as roots. Thus $-6,-1,\frac{3}{2}$ are the roots or solutions of the given equation. ■

3.7.5 Roots in Progressions

As already noted to solve higher degree polynomial equations, we need some information about the solutions of the equation or about the polynomial. "The roots are in arithmetic progression" and "the roots are in geometric progression" are some of such information. Let us discuss an equation of this type.

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